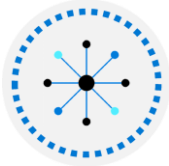


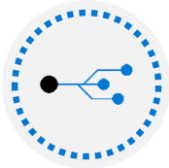
Training Course

Module 04: Virtual Networking

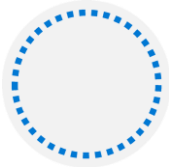




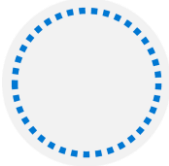
Lesson 01: Configure Virtual Networks



Lesson 02: Configure Network Security Groups

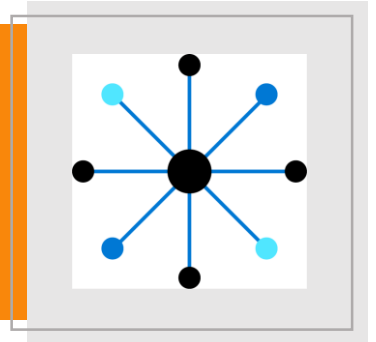


Lesson 03: Configure Azure Firewall

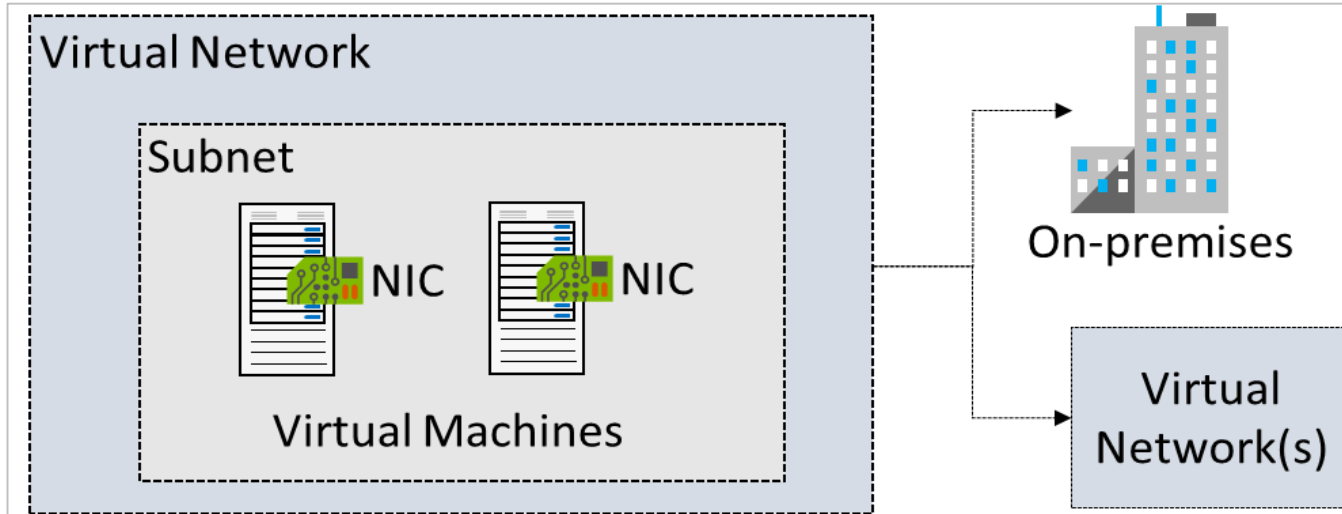


Lesson 04: Configure Azure DNS

Lesson 01: Configure Virtual Networks



Plan Virtual Networks



Logical representation
of your own network

Create a dedicated
private cloud-only
virtual network

Securely extend
your datacenter with
virtual networks

Enable hybrid
cloud scenarios

Create Subnets

+ Subnet	+ Gateway subnet	Refresh	Manage users	Delete
Name ↑↓	IPv4 ↑↓	IPv6 ↑↓	Available IPs ↑↓	Delegated
subnet0	10.0.0.0/24	-	250	-
subnet1	10.0.1.0/24	-	251	-
subnet2	10.0.2.0/24	-	251	-
AzureBastionSubnet	10.0.30.0/27	-	27	-
GatewaySubnet	10.0.3.0/27	-	availability dependent on dynamic use	-

A virtual network can be segmented into one or more subnets

Subnets provide logical divisions within your network

Subnets can help improve security, increase performance, and make it easier to manage the network

Each subnet must have a unique address range – cannot overlap with other subnets in the vnet in the subscription

Create new virtual networks
at any time

Add virtual networks when
you create a virtual machine

Need to define the address space,
and at least one subnet

Be careful with overlapping address
spaces

Create virtual network

Basics IP Addresses Security Tags Review + create

Project details

Subscription * ⓘ

Visual Studio Enterprise



Resource group * ⓘ

Lab04

[Create new](#)

Instance details

Name *

VNet2

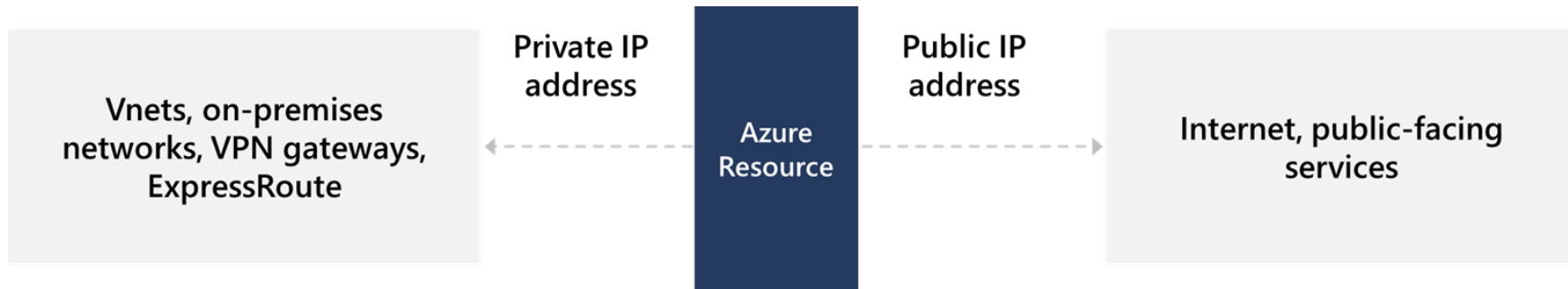


Region *

(US) East US 2



Plan IP Addressing



Private IP addresses – used within an Azure virtual network (VNet), and your on-premises network, when you use a VPN gateway or ExpressRoute circuit to extend your network to Azure

Public IP addresses – used for communication with the Internet, including Azure public-facing services

Available in IPv4 or IPv6 or both

Basic vs Standard SKU

Dynamic vs Static

Zone redundant (Standard SKU)

Range of contiguous addresses available as a prefix

Create public IP address

IP Version * ⓘ

☒ IPv4 ☐ IPv6 ☐ Both

SKU * ⓘ

☒ Basic ☐ Standard

IPv4 IP Address Configuration

Name *

IP address assignment *

☒ Dynamic ☐ Static

Associate Public IP Addresses

Public IP addresses	IP address association	Dynamic	Static
Virtual Machine	NIC	Yes	Yes
Load Balancer	Front-end configuration	Yes	Yes
VPN Gateway	Gateway IP configuration	Yes	Yes*
Application Gateway	Front-end configuration	Yes	Yes*

A public IP address resource can be associated with virtual machine network interfaces, internet-facing load balancers, VPN gateways, and application gateways

*Static IP addresses only available on certain SKUs.

Associate Private IP Addresses

Private IP Addresses	IP address association	Dynamic	Static
Virtual Machine	NIC	Yes	Yes
Internal Load Balancer	Front-end configuration	Yes	Yes
Application Gateway	Front-end configuration	Yes	Yes

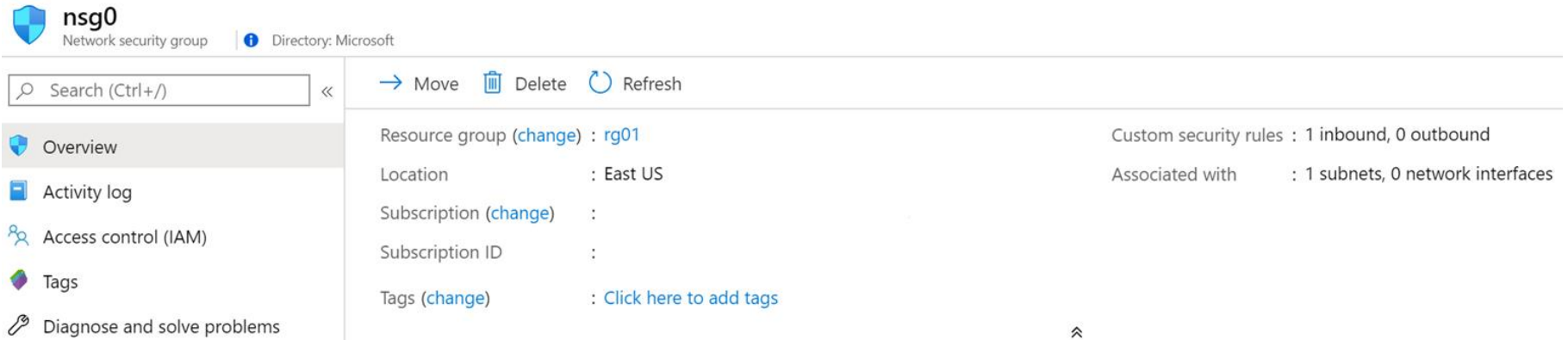
Dynamic (default) Azure assigns the next available unassigned or unreserved IP address in the subnet's address range

Static You select and assign any unassigned or unreserved IP address in the subnet's address range

Lesson 02: Configure Network Security Groups



Implement Network Security Groups



The screenshot shows the Azure portal interface for a Network Security Group (nsg0). The left sidebar contains navigation links: Overview (selected), Activity log, Access control (IAM), Tags, and Diagnose and solve problems. The main content area displays the following details:

- Resource group (change) : rg01
- Location : East US
- Subscription (change) :
- Subscription ID :
- Tags (change) : [Click here to add tags](#)
- Custom security rules : 1 inbound, 0 outbound
- Associated with : 1 subnets, 0 network interfaces

Limits network traffic to resources in a virtual network

Lists the security rules that allow or deny inbound or outbound network traffic

Associated to a subnet or a network interface

Can be associated multiple times

Determine NSG Rules

Inbound security rules

Priority	Name	Port	Protocol	Source	Destination	Action
100	 RDP_Inbound	3389	Any	Any	Any	 Allow
65000	AllowVnetInBound	Any	Any	VirtualNetwork	VirtualNetwork	 Allow
65001	AllowAzureLoadBalancerInBound	Any	Any	AzureLoadBalancer	Any	 Allow
65500	DenyAllInBound	Any	Any	Any	Any	 Deny

Outbound security rules

Priority	Name	Port	Protocol	Source	Destination	Action
65000	AllowVnetOutBound	Any	Any	VirtualNetwork	VirtualNetwork	 Allow
65001	AllowInternetOutBound	Any	Any	Any	Internet	 Allow
65500	DenyAllOutBound	Any	Any	Any	Any	 Deny

Security rules in NSGs enable you to filter network traffic that can flow in and out of virtual network subnets and network interfaces

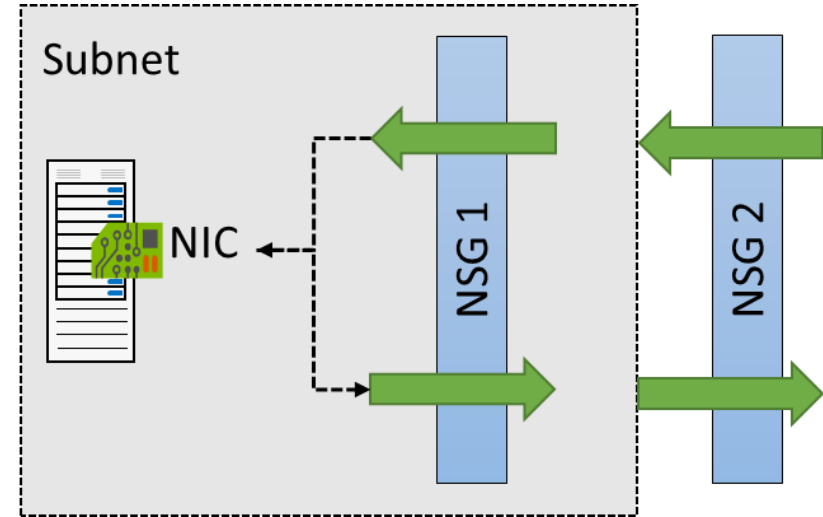
There are default security rules. You cannot delete the default rules, but you can add other rules with a higher priority

Determine NSG Effective Rules

NSGs are evaluated independently for the subnet and NIC

An “allow” rule must exist at both levels for traffic to be admitted

Use the Effective Rules link if you are not sure which security rules are being applied



Network Interface: [vm01990](#) [Effective security rules](#) [Topology](#)
Virtual network/subnet: [vnet01/subnet0](#) NIC Public IP: - NIC Private IP: **10.1.0.4** Accelerated networking: **Disabled**

Create NSG rules

Select from a large variety of services

Service – The destination protocol and port range for this rule

Port ranges – Single port or multiple ports

Priority – The lower the number, the higher the priority

The screenshot shows the 'Add inbound security rule' dialog box for a network security group named 'nsg1'. The 'Advanced' tab is selected. The fields are as follows:

- Service**: A dropdown menu showing 'Custom'.
- Port ranges**: A text input field containing '8080'.
- Priority**: A text input field containing '100'.
- Name**: A text input field containing 'Port_8080'.
- Description**: An empty text area.

Lesson 03: Configure Azure Firewall



Determine Azure Firewall Uses

Stateful firewall as a service

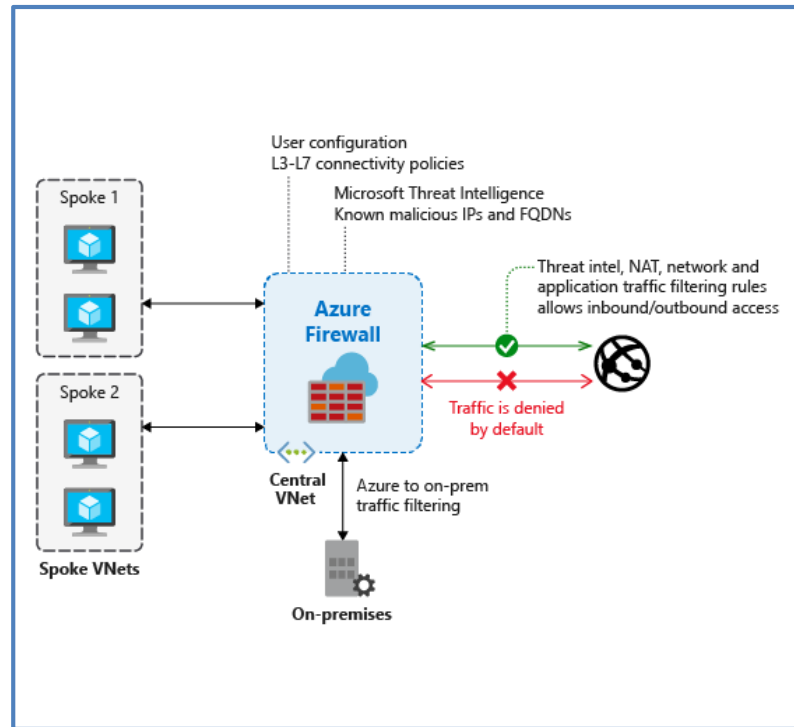
Built-in high availability with unrestricted cloud scalability

Create, enforce, and log application and network connectivity policies

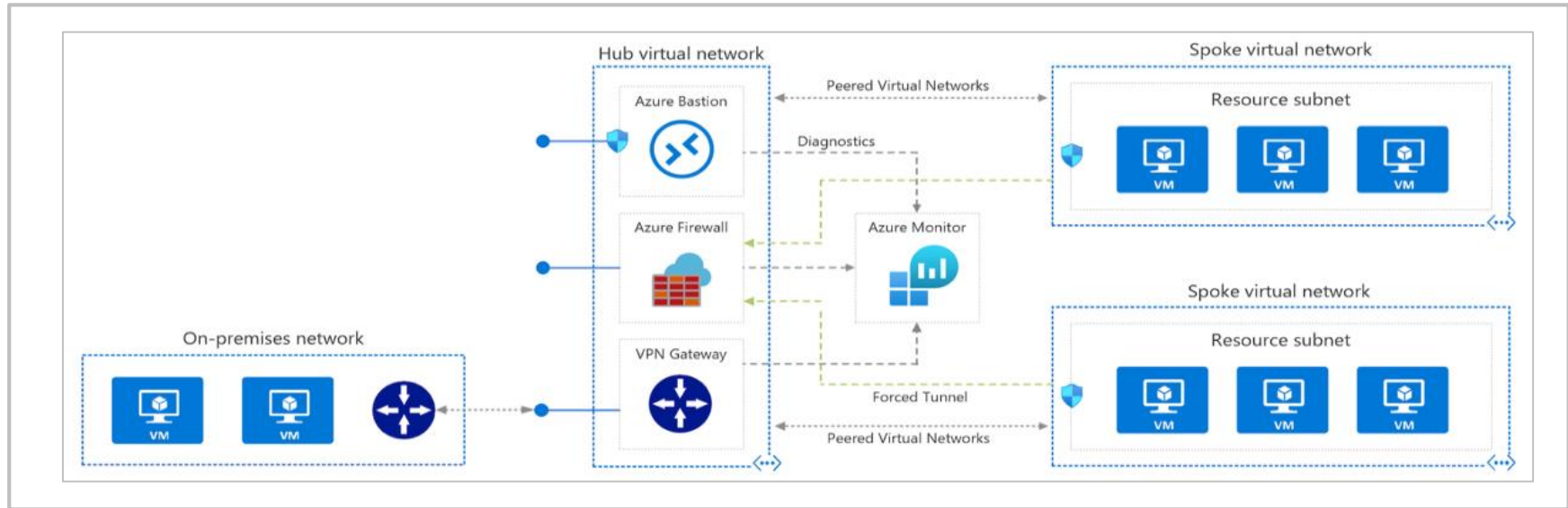
Threat intelligence-based filtering

Fully integrated with Azure Monitor for logging and analytics

Support for hybrid connectivity through deployment behind VPN and ExpressRoute Gateways



Create Azure Firewalls

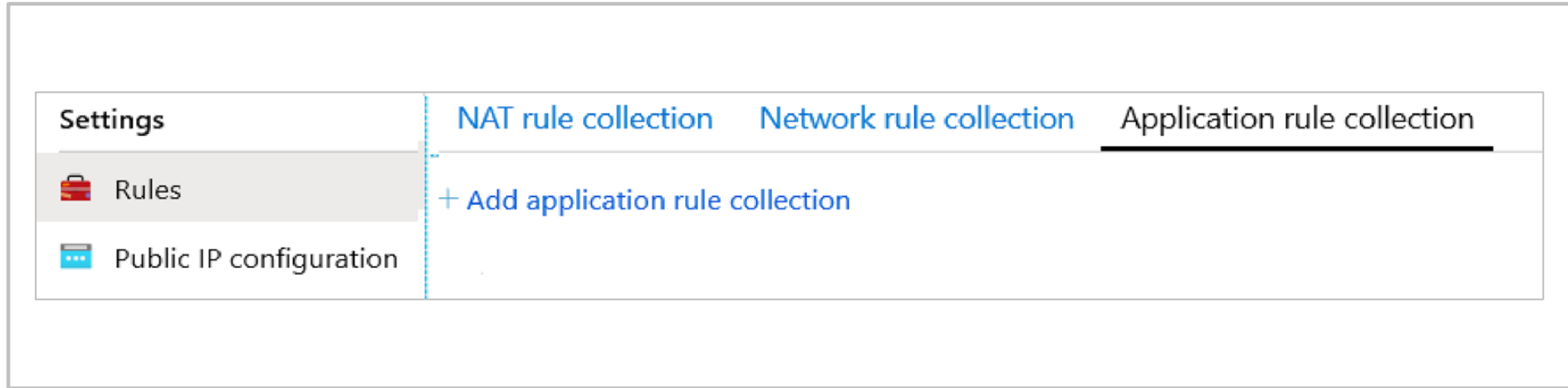


A Hub-Spoke network topology is recommended

Shared services are placed in the hub virtual network

Each environment is deployed to a spoke to maintain isolation

Create Azure Firewall Rules

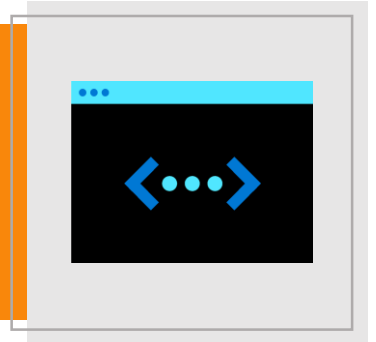


NAT rules. Configure DNAT rules to allow incoming connections

Network rules. Configure rules that contain source addresses, protocols, destination ports, and destination addresses

Application rules. Configure fully qualified domain names (FQDNs) that can be accessed from a subnet

Lesson 04: Configure Azure DNS



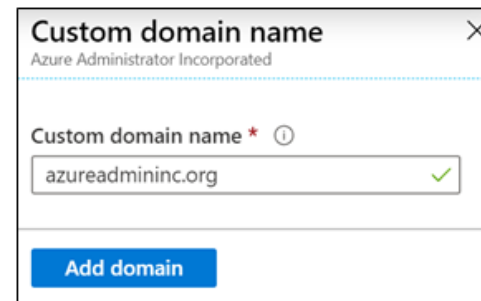
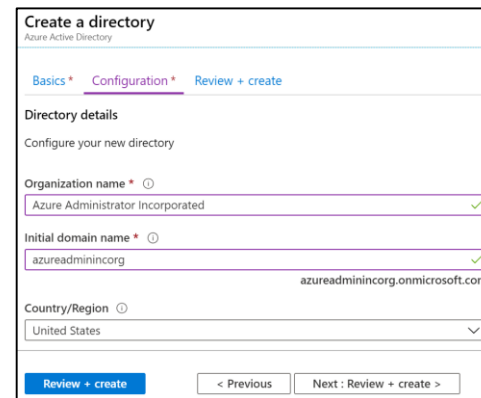
Identity Domains and Custom Domains

When you create an Azure subscription an Azure AD domain is created for you

The domain has initial domain name in the form *domainname.onmicrosoft.com*

You can customize/change the name

After the custom name is added it must be verified (next topic)



Verify Custom Domain Names

Verification demonstrates ownership of the domain name

Add a DNS record (MX or TXT) that is provided by Azure into your company's DNS zone

Azure will query the DNS domain for the presence of the record

This could take several minutes or several hours

azureadmininc.org

Custom domain name

 Delete |  Got feedback?

 To use azureadmininc.org with your Azure AD, create a new TXT record with your domain name registrar using the info below.

Record type

TXT

☐ MX

Alias or host name

@

Destination or points to address

MS=ms79094380

TTL

3600

[Share these settings via email](#)

Verification will not succeed until you have configured your domain with your registrar as described above.

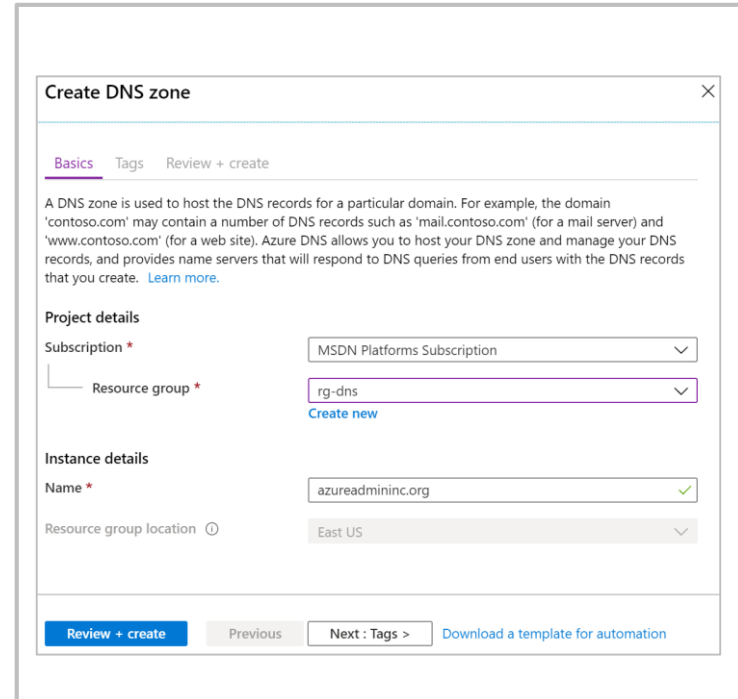
Create Azure DNS Zones

A DNS zone hosts the DNS records for a domain

The name of the zone must be unique within the resource group

Where multiple zones share the same name, each instance is assigned different name server addresses

Root/Parent domain is registered at the registrar and pointed to Azure NS



The screenshot shows the 'Create DNS zone' wizard in the Azure portal. The wizard has three tabs: 'Basics' (selected), 'Tags', and 'Review + create'. Below the tabs, there is a descriptive paragraph about DNS zones. The 'Project details' section includes a 'Subscription' dropdown set to 'MSDN Platforms Subscription' and a 'Resource group' dropdown set to 'rg-dns', with a 'Create new' link below it. The 'Instance details' section includes a 'Name' field with 'azureadmininc.org' and a green checkmark, and a 'Resource group location' dropdown set to 'East US'. At the bottom, there are buttons for 'Review + create', 'Previous', 'Next : Tags >', and a link to 'Download a template for automation'.

Create DNS zone

Basics Tags Review + create

A DNS zone is used to host the DNS records for a particular domain. For example, the domain 'contoso.com' may contain a number of DNS records such as 'mail.contoso.com' (for a mail server) and 'www.contoso.com' (for a web site). Azure DNS allows you to host your DNS zone and manage your DNS records, and provides name servers that will respond to DNS queries from end users with the DNS records that you create. [Learn more.](#)

Project details

Subscription * MSDN Platforms Subscription

Resource group * rg-dns [Create new](#)

Instance details

Name * azureadmininc.org

Resource group location East US

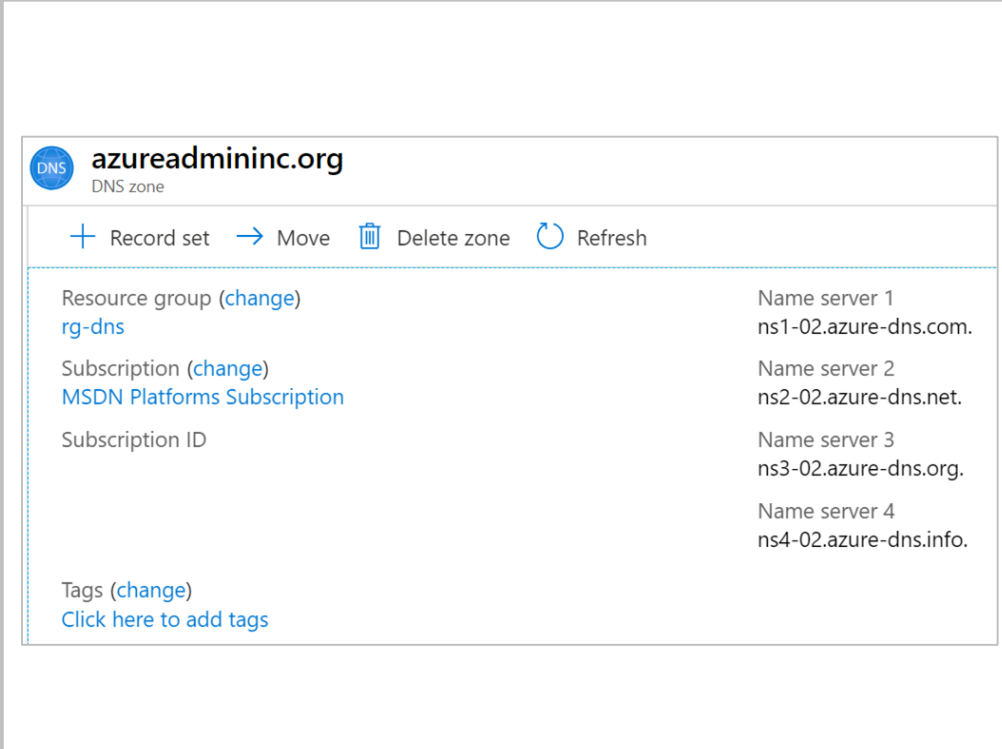
[Review + create](#) [Previous](#) [Next : Tags >](#) [Download a template for automation](#)

Delegate DNS Domains

When delegating a domain to Azure DNS, you must use the name server names provided by Azure DNS – use all four

Once the DNS zone is created, update the parent registrar

For child zones, register the NS records in the parent domain



The screenshot displays the Azure portal interface for a DNS zone named 'azureadmininc.org'. At the top, there's a header with the domain name and a 'DNS zone' label. Below this, a toolbar contains icons for '+ Record set', 'Move', 'Delete zone', and 'Refresh'. The main content area is divided into two columns. The left column lists metadata: 'Resource group (change) rg-dns', 'Subscription (change) MSDN Platforms Subscription', and 'Subscription ID'. The right column lists the four required name servers: 'Name server 1 ns1-02.azure-dns.com.', 'Name server 2 ns2-02.azure-dns.net.', 'Name server 3 ns3-02.azure-dns.org.', and 'Name server 4 ns4-02.azure-dns.info.'. At the bottom left, there are links for 'Tags (change)' and 'Click here to add tags'.

azureadmininc.org DNS zone	
+ Record set → Move Delete zone Refresh	
Resource group (change) rg-dns	Name server 1 ns1-02.azure-dns.com.
Subscription (change) MSDN Platforms Subscription	Name server 2 ns2-02.azure-dns.net.
Subscription ID	Name server 3 ns3-02.azure-dns.org.
	Name server 4 ns4-02.azure-dns.info.
Tags (change) Click here to add tags	

Add DNS Record Sets

A record set is a collection of records in a zone that have the same name and are the same type

You can add up to 20 records to any record set

A record set cannot contain two identical records

Changing the drop-down Type, changes the information required

Add record set

azureadmininc.org

Name

helloworld ✓

.azureadmininc.org

Type

A

Alias record set ⓘ

☐ Yes ☒ No

TTL * TTL unit

IP address

0.0.0.0 ...

Plan for Private DNS Zones

Use your own custom domain names

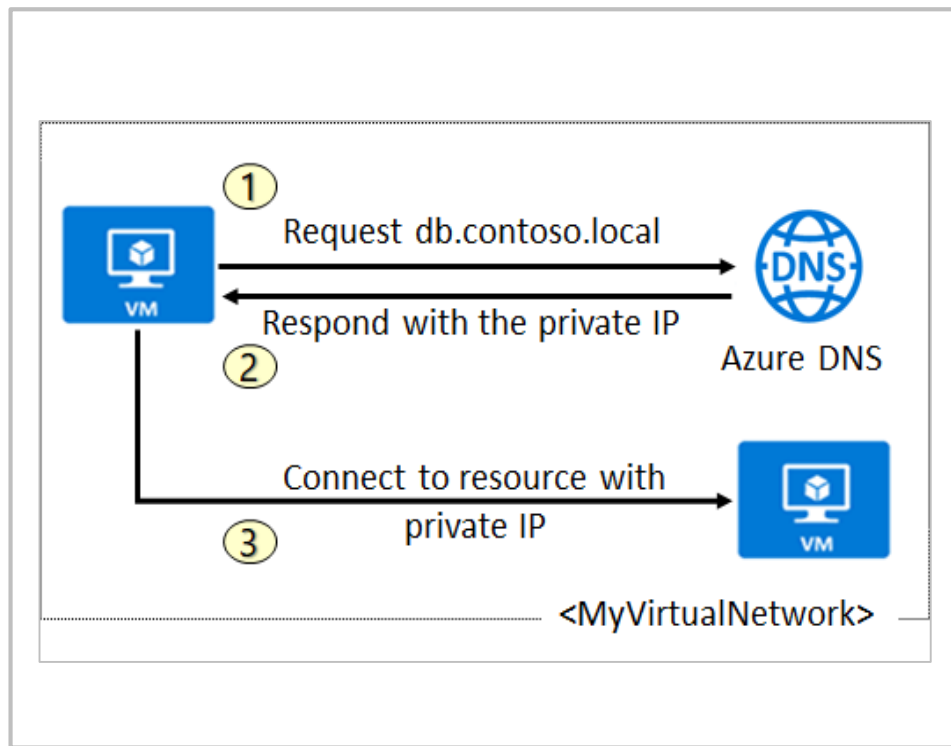
Provides name resolution for VMs within a VNet and between VNets

Automatic hostname record management

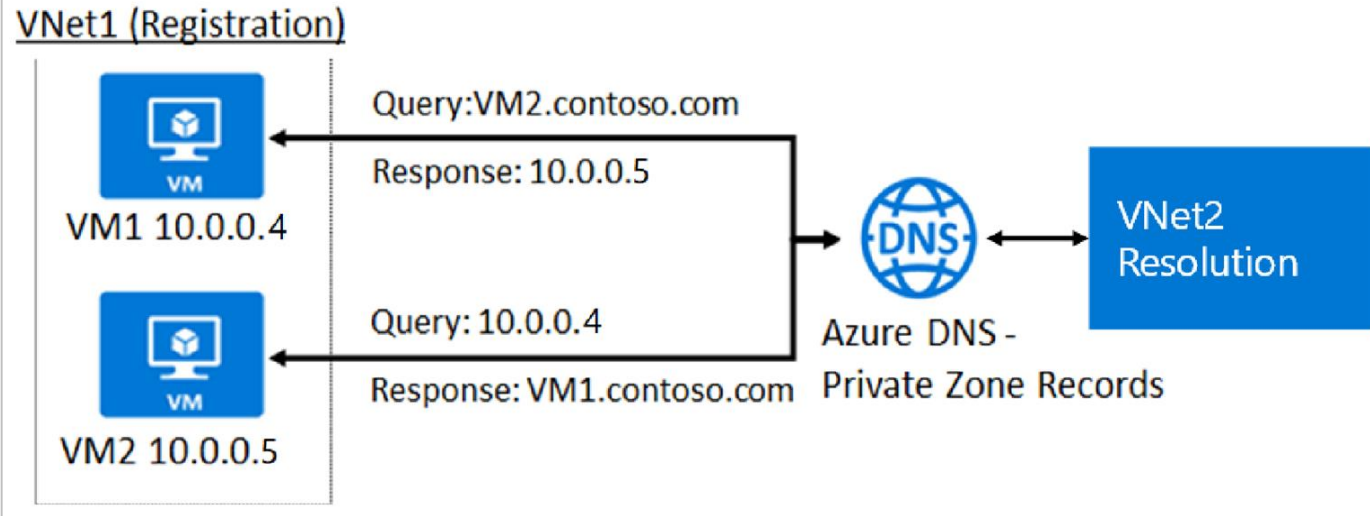
Removes the need for custom DNS solutions

Use all common DNS records types

Available in all Azure regions



Determine Private Zone Scenarios



DNS resolution in VNet1 is private and not accessible from the Internet

DNS queries across the virtual networks are resolved

Reverse DNS queries are scoped to the same virtual network

Thank you

